

FULL NAME: _____

BLOCK: _____

Logarithms + DOK-3 Spine: q15 (exp $\leftrightarrow$ log inverse via graph)

Lesson 6-3 · Quick · Student Packet

OBJECTIVES

Math Objective	Evaluate logarithms using the definition $\log_b x = y \Leftrightarrow b^y = x$, distinguish common (log) vs natural (ln) logs, and use the graph of $y = 3^x$ to estimate logarithmic values through the exp $\leftrightarrow$ log inverse relationship.
Language Objective	Explain the inverse relationship between exponential and logarithmic equations using “undo” language.
Essential Understanding	A logarithm answers the question: what exponent? – logs undo exponentials. If I know how to graph $y = 3^x$, I can use that same picture to estimate $\log_3 50$ because logarithms and exponentials are inverses.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency evaluating logarithms; use the exp $\leftrightarrow$ log inverse relationship graphically (Topic 6 Form B prep: q15 covers Q7, Q10, Q11; q17 covers Q13).
Essential Question	If I know how to graph $y = 3^x$, how can I use that same picture to estimate $\log_3 50$ — and why does this work?
Materials	Student packet with printed $y = 3^x$ graph ($x \in [-1, 4.5]$, $y \in [0, 85]$, grid-lines). Teacher copy adds dashed $y = 50$ line and read-off annotation. [GRAPH PLACEHOLDER — needs_cleanup=YES, see Visuals Checklist].

LAUNCH — Evaluate Logarithms (teacher think-aloud Ex 3)**[BOOK] LOG $\leftrightarrow$ EXP RULES (keep printed on your page)**

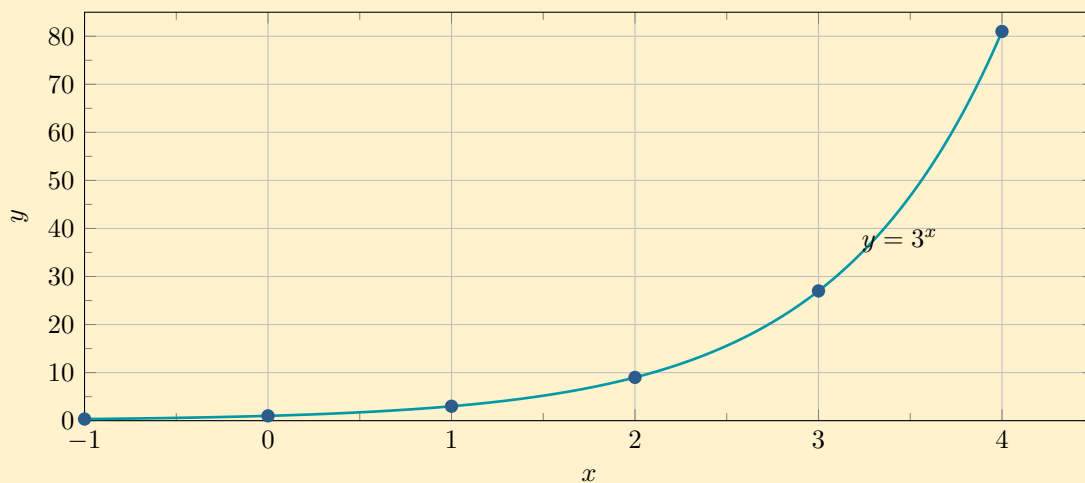
LOG $\Leftrightarrow$ EXP: $\log_b x = y \Leftrightarrow b^y = x$
 COMMON LOG: $\log x = \log_{10} x$
 NATURAL LOG: $\ln x = \log_e x$
 KEY: $\ln(e^t) = t$ (this solves continuous growth!)
 UNDEFINED: $\log_b 0$ and $\log_b(\text{negative})$ — no real answer
 IDENTITY: $\log_b(b^k) = k$

EXPLORE — Think-Pair-Share (25 min)

Work with a partner. Start with [STAR] Practice #15 (DOK-3 spine — 10 min TPS + 5 min pair presentations). Then move to #52, #53, #54, and #17.

[GRAPH] GRAPH FOR q15 — $y = 3^x$ [GRAPH PLACEHOLDER]

[GRAPH PLACEHOLDER — print the $y = 3^x$ graph here: $x \in [-1, 4.5]$, $y \in [0, 85]$, with gridlines. Source: a2_6-3_SE.pdf near end of practice problems. See Visuals Checklist — needs_cleanup=YES.]



Bank-item labels (in order):

1. Practice #15 [STAR] DOK-3 SPINE — Higher Order Thinking: Use $y = 3^x$ graph to estimate $\log_3 50$
2. Practice #52 — DOK-2 Application
3. Practice #53 — DOK-2 Application [45-min variant: skip]
4. Practice #54 — DOK-2 Application [45-min variant: skip]
5. Practice #17 — DOK-3 Proof-style: critique $\log_3 \left(\frac{1}{27}\right) = -3$ (supports Form B Q13)

Practice #15 [STAR] DOK-3 SPINE — Higher Order Thinking: Use $y = 3^x$ graph to estimate $\log_3 50$

Higher Order Thinking Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.
[GRAPH / TIKZ figure]

Practice #52 — DOK-2 Application

How long does it take for \$250 to grow to \$600 at 4% annual percentage rate compounded continuously? Round to the nearest year.

Practice #53 — DOK-2 Application [45-min variant: skip]

Model with Mathematics Michael invests \$1,000 in an account that earns a 4.75% annual percentage rate compounded continuously. Peter invests \$1,200 in an account that earns a 4.25% annual percentage rate compounded continuously. Which person's account will grow to \$1,800 first?

Practice #54 — DOK-2 Application [45-min variant: skip]

Reason The Richter magnitude of an earthquake is $R = 0.67 \log(0.37E) + 1.46$, where E is the energy (in kilowatt-hours) released by the earthquake.

[IMAGE: Cracked wall photo with Richter scale overlay numbered 1 through 10 and labels “Not felt” at 1 and “Extreme” near 9–10; callout text: “At a richter magnitude of 4 and above, the walls in your house may start to crack.”]

- a. What is the magnitude of an earthquake that releases 11,800,000,000 kilowatt-hours of energy? Round to the nearest tenth.

- b. How many kilowatt-hours of energy would earthquake have to release in order to be an 8.2 on the Richter scale? Round to the nearest whole number.
- c. What number of kilowatt-hours of energy would an earthquake have to release in order for walls to crack? Round to the nearest whole number.

Practice #17 — DOK-3 Proof-style: critique $\log_3 \left(\frac{1}{27}\right) = -3$ (supports Form B Q13)

Use Structure A student says that $\log_3 \left(\frac{1}{27}\right)$ simplifies to -3. Is the student correct? Explain.

[NOTE] LOG $\leftrightarrow$ EXP RULES — Reference Card (print this!)

LOG $\Leftrightarrow$ EXP: $\log_b x = y \Leftrightarrow b^y = x$

COMMON LOG: $\log x = \log_{10} x$

NATURAL LOG: $\ln x = \log_e x$

KEY: $\ln(e^t) = t$ (this solves continuous growth!)

UNDEFINED: $\log_b 0$ and $\log_b(\text{negative})$ — no real answer

IDENTITY: $\log_b(b^k) = k$

SHARE / SUMMARY (5 min)

Sentence frames

Pair share (q15): On the graph of $y = 3^x$, I found $\log_3 50$ by looking for $y = \underline{\hspace{2cm}}$ and reading the x-value. Since $3^3 = \underline{\hspace{2cm}}$ and $3^4 = \underline{\hspace{2cm}}$, the answer is between $\underline{\hspace{2cm}}$ and $\underline{\hspace{2cm}}$, closer to $\underline{\hspace{2cm}}$. This works because $\log_3 50 = x$ means $3^x = \underline{\hspace{2cm}}$, so the exponential and logarithm are $\underline{\hspace{2cm}}$.

Whole class: one pair explains why reading the graph works. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO LESSON 6-4

Lesson 6-4 shifts to logarithmic FUNCTIONS — graphing $y = \log_b x$, identifying domain, range, and asymptotes. Today's log $\leftrightarrow$ exp definition is the foundation.

EXIT

$\log_b x = y$ means $\underline{\hspace{2cm}}$.

$\log_2 32 = \underline{\hspace{2cm}}$ because $2^? = 32$.

To solve $e^t = 6.125$, take the $\underline{\hspace{2cm}}$ of both sides giving $t = \underline{\hspace{2cm}}$.

One biggest thing I learned today: $\underline{\hspace{2cm}}$

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #18 asks you to evaluate whether $\ln 1000 = 3$ — use the $\log \leftrightarrow \exp$ definition and the value of e to verify or refute.

Optional #1 (Savvas Practice #18 — $\ln 1000 \neq 3$?)

Use Structure Explain why the expression $\ln 1,000$ is not equal to 3.